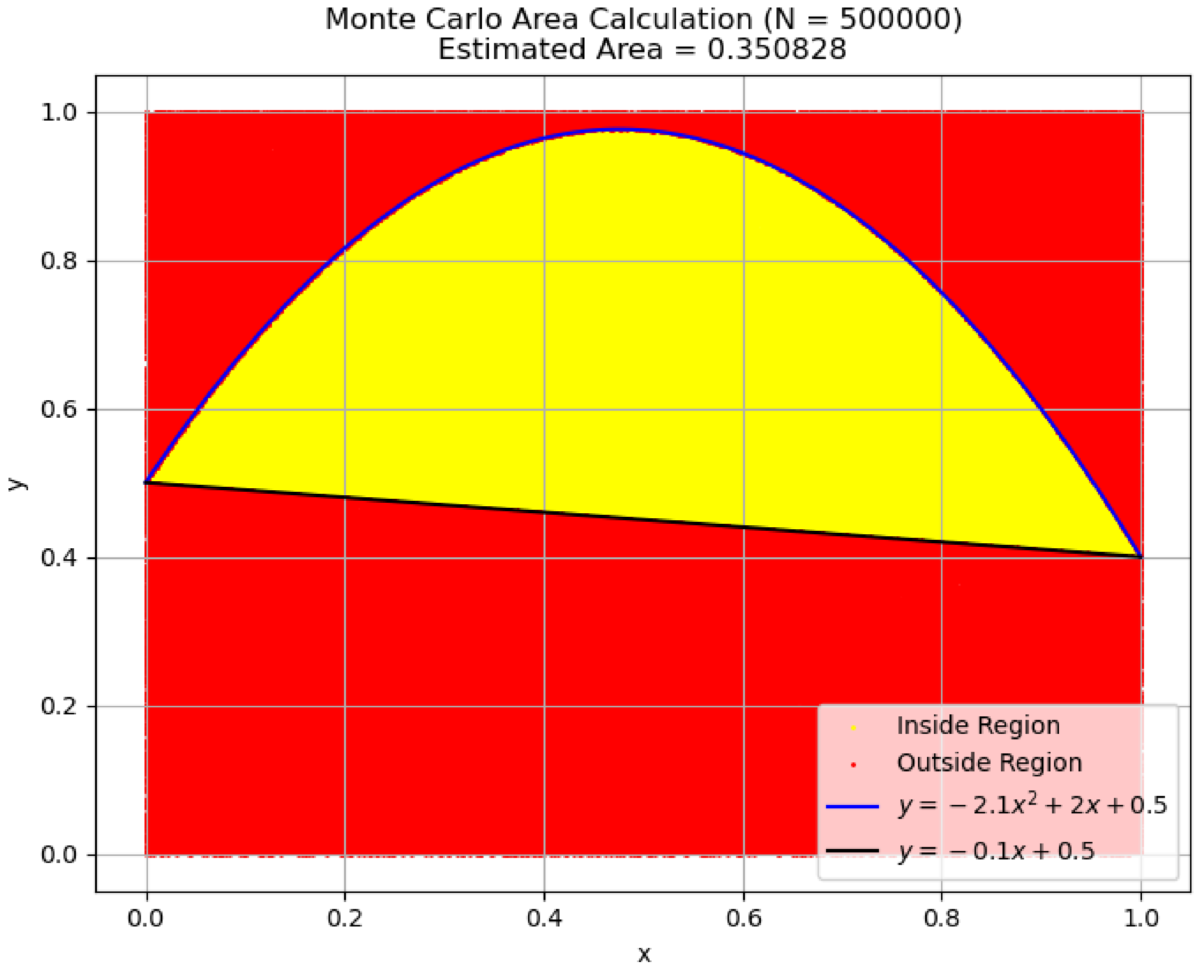
In [5]:

```
import random
import matplotlib.pyplot as plt

def f_upper(x):
    return -2.1 * x**2 + 2 * x + 0.5

def f_lower(x):
    return -0.1 * x + 0.5

def monte_carlo_area(num_points):

    points_in_region = 0
    x_in_region = []
    y_in_region = []
    x_outside_region = []
    y_outside_region = []

    x_min, x_max = 0, 1
    y_min, y_max = 0, 1
    area_of_box = (x_max - x_min) * (y_max - y_min)

    for _ in range(num_points):
        x = random.uniform(x_min, x_max)
        y = random.uniform(y_min, y_max)

        if f_lower(x) <= y <= f_upper(x):
            points_in_region += 1
            x_in_region.append(x)
            y_in_region.append(y)
        else:
            x_outside_region.append(x)
            y_outside_region.append(y)

    estimated_area = area_of_box * (points_in_region / num_points)

    plt.figure(figsize=(8, 6))

    plt.scatter(x_in_region, y_in_region, color='yellow', s=1, label='Inside Region
    plt.scatter(x_outside_region, y_outside_region, color='red', s=1, label='Outsid

    x_values = [i / 100 for i in range(101)]
    plt.plot(x_values, [f_upper(x) for x in x_values], color='blue', label='$y = -2
    plt.plot(x_values, [f_lower(x) for x in x_values], color='black', label='$y = -

    plt.title(f'Monte Carlo Area Calculation (N = {num_points})\nEstimated Area = {
    plt.xlabel('x')
    plt.ylabel('y')
    plt.legend()
    plt.grid(True)
    plt.show()

    return estimated_area

if __name__ == '__main__':
```

```python
    num_points = 500000
    area = monte_carlo_area(num_points)

    print(f"The estimated area of the region is: {area:.6f}")
```




```
The estimated area of the region is: 0.350828
```

In [ ]: